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| <b>PROJECT:</b> | MOS Ref No:          | MOS/ACMV/WEPL/<br>RPT-01 |
|                 | Revision No:         | 02                       |
|                 | Total No. of<br>Page | 15                       |

## METHOD OF STATEMENT FOR ROOM PRESSURE TESTING (POSITIVE & NEGATIVE) FOR ACMV SYSTEMS

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## **1.0 Objective**

To define the systematic procedure for the testing, verification, and documentation of room pressure differentials (both positive and negative) for specialized environments. This procedure ensures that the performance of the Air-Conditioning and Mechanical Ventilation (ACMV) system meets the design intent, project specifications, and the principles of relevant project standard, thereby guaranteeing operational integrity, safety, and contamination control.

## **2.0 Scope of Works**

This Method of Statement covers all activities related to the on-site measurement and verification of room pressurization. This includes pre-test checks, equipment setup, execution of both quantitative (micromanometer or TSI Multi-function meter) and qualitative (smoke visualization) tests, data recording, and reporting for designated rooms within the project. This procedure is applicable after the mechanical completion and pre-commissioning of the ACMV system.

## **3.0 References**

- Practice for Air-Conditioning and Mechanical Ventilation in Buildings.
- Practice for Indoor Air Quality for Air-Conditioned Buildings.
- **Workplace Safety and Health Act (WSHA):** And associated regulations.
- **Project-Specific Documents:** Including technical specifications and approved design drawings.
- **Approved Testing, Adjusting, and Balancing (TAB) Plan.**
- **Equipment Manufacturer's Installation.**

## **4.0 Responsibilities**

- **Project Manager:** Holds overall responsibility for the successful and safe execution of the testing works, ensuring allocation of necessary resources.
- **Project Coordinator:** Responsible for supervising the testing procedure, interpreting the results against design specifications, witnessing the tests, and formally approving the test reports.
- **Technician:** Responsible for executing the tests as per this method statement, operating the test equipment, and collecting accurate data.
- **Safety Personal:** Responsible for ensuring all work is conducted safely, all personnel have the required Personal Protective Equipment (PPE), and a risk assessment is in place.

## 5.0 Equipment and Tools

- **Calibrated Digital Micromanometer:** A high-precision instrument capable of measuring low-pressure differentials (e.g., in Pascals). The device must have a valid calibration certificate.
- **Pressure Probes and Tubing:** Appropriate lengths of flexible tubing to connect the micromanometer from the test room to the reference area without being pinched.
- **Non-Toxic Smoke Generator / Smoke Puffer / Smoke Tubes:** For qualitative airflow visualization. The smoke should be non-staining and safe for inhalation in small quantities.
- **Display Monitor (Optional):** For continuous monitoring to verify pressure stability over a set period in critical applications.
- **Warning Signs & Barriers:** To secure the test area (e.g., "TESTING IN PROGRESS - DO NOT ENTER").
- **Personal Protective Equipment (PPE):** Safety glasses, gloves, and any other site-specific requirements.

|                                                                                                                         |                                                                                                                                |                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|  <p>Digital multi-function meter</p> |  <p>Pressure Probes and Tubing</p>          |  <p>Non-Toxic Smoke Generator</p>    |
|  <p>Display Monitor (Optional)</p>   |  <p>Personal Protective Equipment (PPE)</p> |  <p>Warning Signs &amp; Barriers</p> |

### **6.1. Documentation and System Review**

- Obtain the final approved architectural and ACMV drawings.
- Identify all rooms requiring pressure testing and note the specified pressure differential value (e.g., +15 Pa, -10 Pa) relative to their designated reference area (e.g., anteroom, corridor).
- Confirm that the Testing, Adjusting, and Balancing of airflows (Supply, Return, Exhaust) for the relevant zones has been completed and signed off. The pressure differential is a direct result of the airflow balance.

### **6.2. Verification of Room Envelope Integrity**

- Conduct a thorough visual inspection of the room to be tested. All six surfaces (floor, ceiling, four walls) must be structurally complete.
- **Seal all penetrations:** Check that all openings for pipes, conduits, ducts, and other services passing through the walls or ceiling are meticulously sealed with appropriate sealant (e.g., fire-rated mastic, silicon).
- **Inspect Doors:** Ensure doors are correctly installed, not warped, and close properly against their frames. Check that all door gaskets and sweeps are in place and in good condition.
- **Inspect Windows and Glazing:** Verify that all windows are fully closed and that their seals are intact.
- **Ceiling Tiles:** For rooms with suspended ceilings, ensure all ceiling tiles are properly seated in the grid.

### **6.3. ACMV System Operation**

- Ensure the ACMV system serving the room and adjacent areas is running under its normal operational mode for at least 30 minutes to allow conditions to stabilize.
- Verify with the Building Management System (BMS) or controls contractor that all associated VAV boxes, dampers, and fans are operating at their design setpoints.
- Close all doors and windows to the room being tested and to the adjacent reference areas.

## **7.0 Detailed Testing Procedure**

### **7.1. General Instrument Setup**

1. Power on the digital micromanometer or TSI Multi-function meter and allow it to perform its self-zeroing calibration sequence in a neutral space.
2. Connect the flexible tubing to the "high pressure" (+) port and the "low pressure" (-) port of the device.
3. Designate the room to be tested as the "controlled space" and the adjacent corridor or anteroom as the "reference space."

### **7.2. Procedure for POSITIVE Pressure Room Testing**

A positive pressure room pushes air out, preventing contaminants from entering.

#### **Step 1: Quantitative Measurement (The Number)**

1. Carefully pass the tubing from the **low-pressure (-)** port under the door of the controlled space into the adjacent reference space. Ensure the tube is not kinked or flattened by the door.
2. The tube connected to the **high-pressure (+)** port remains inside the controlled (positive pressure) room.
3. Position the tips of both tubes away from direct airflow from diffusers, exhaust grilles, or doorways to avoid velocity-induced pressure errors.
4. Allow the reading on the micromanometer to stabilize. This may take 30-60 seconds.
5. Record the stabilized pressure differential. The reading **must be a positive value** (e.g., +12.5 Pa), confirming the room's pressure is higher than the reference space.
6. Record the reading on the official Test Record Sheet.

#### **Step 2: Qualitative Verification (The Visual Proof)**

1. Have a second person stand outside the door in the reference area.
2. Slightly crack the door open (approx. 20-30mm).
3. From the outside, generate a small puff of smoke near the bottom of the door gap.

4. **Observe the smoke's direction.** For a correctly pressurized positive room, the smoke will be actively **pushed away from the door gap** and out into the reference space.
5. Document this observation with a photograph for the report.

### **7.3. Procedure for NEGATIVE Pressure Room Testing**

A negative pressure room pulls air in, containing contaminants within the room.

#### **Step 1: Quantitative Measurement (The Number)**

1. The instrument setup is identical to the positive pressure test. Pass the tubing from the **low-pressure (-)** port under the door into the adjacent reference space.
2. The tube connected to the **high-pressure (+)** port remains inside the controlled (negative pressure) room.
3. Allow the reading on the micromanometer or TSI Multi-function meter to stabilize.
4. Record the stabilized pressure differential. The reading **must be a negative value** (e.g., -8.0 Pa), confirming the room's pressure is lower than the reference space.
5. Record the reading on the official Test Record Sheet.

#### **Step 2: Qualitative Verification (The Visual Proof)**

1. Slightly crack the door open (approx. 20-30mm).
2. From the outside (in the reference space), generate a small puff of smoke near the bottom of the door gap.
3. **Observe the smoke's direction.** For a correctly pressurized negative room, the smoke will be actively **sucked into the room** through the door gap.
4. Document this observation with a photograph or short video clip.

### **8.0 Acceptance Criteria**

1. **Quantitative Pass:** The recorded pressure differential from the micromanometer or TSI Multi-function meter must be within the tolerance specified in the project design documents (e.g., Design = -10 Pa; Acceptance Range = -9 Pa to -11 Pa).
2. **Qualitative Pass:** The smoke visualization test must unambiguously confirm the correct direction of airflow (out for positive, in for negative).

3. **Stability:** The pressure reading must remain stable and not fluctuate erratically, which could indicate unstable fan operation or external influences like wind.

## **9.0 Post-Testing and Reporting**

### **9.1. Corrective Actions (If Test Fails)**

- If the measured pressure does not meet the criteria, the contractor will investigate. The investigation will follow a logical sequence:
  1. **Re-check Air Balance:** Verify the supply, return, and exhaust CMH values. Adjust fan speeds or dampers as necessary. This is the most common cause of failure.
  2. **Inspect for Leakage:** If air balance is correct, conduct a detailed search for air leaks in the room envelope. Check sealant around penetrations, door gaskets, and ceiling-to-wall junctions.
- After any corrective actions are implemented, the room must be formally re-tested by repeating the procedures in Section 7.0.

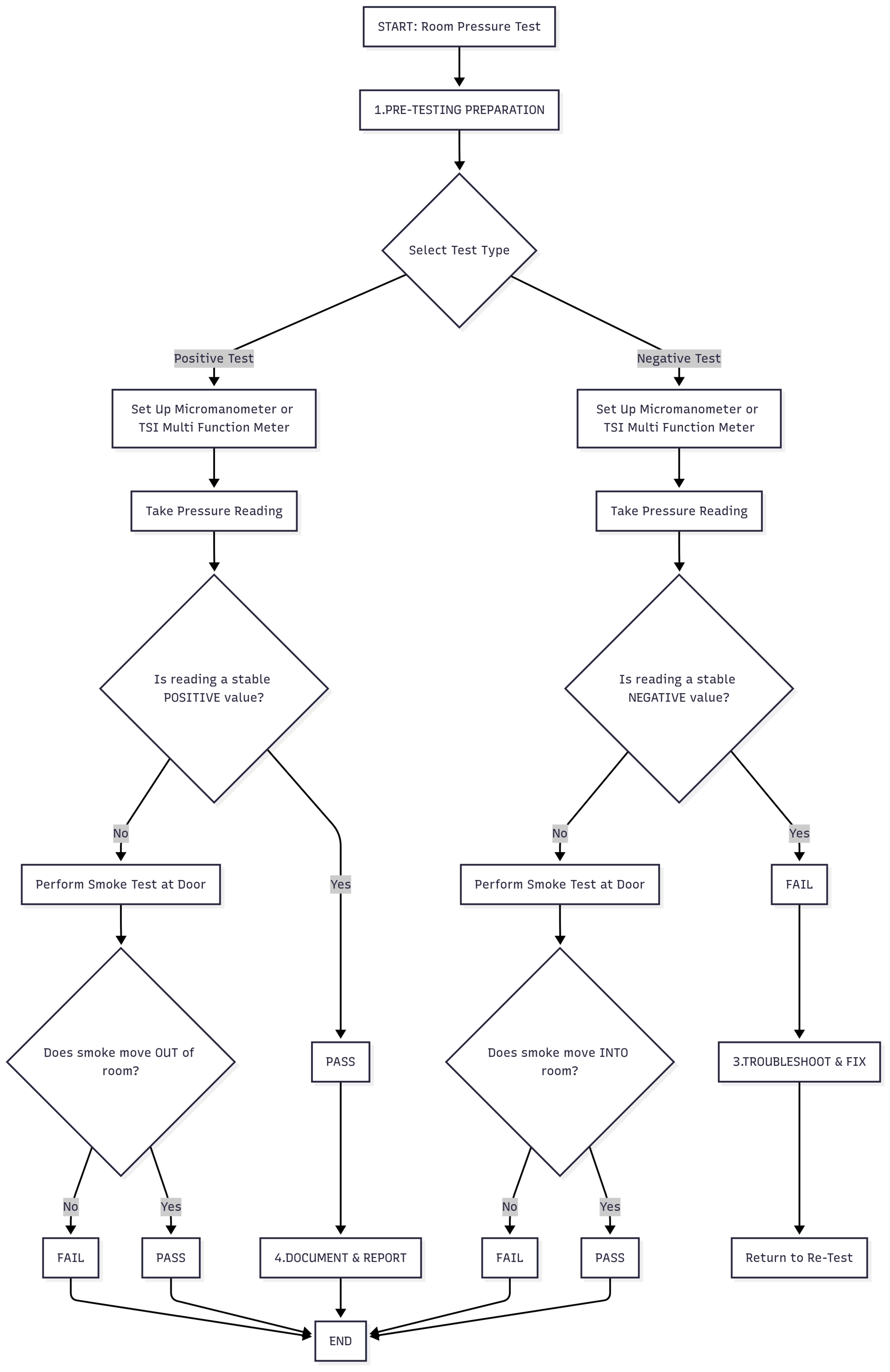
### **9.2. Final Report**

- Upon successful completion, a formal Test Report shall be submitted for each tested room. The report must include:
  - Project Name and Location
  - Date and Time of Test
  - Room Number / Identification
  - ACMV System ID serving the room
  - Specified Design Pressure Differential
  - Measured Pressure Differential
  - Result of Smoke Visualization Test (with photo/video evidence attached)
  - Details of Test Instrument Used (Model, Serial No., Calibration Due Date)
  - Names and Signatures of the testing personnel and the witnessing M&E Engineer.

**10.0 Health, Safety & Environment (HSE)**

- All personnel must have completed a site safety induction and be aware of the project's risk assessments.
- Full PPE (safety shoes, glasses, etc.) is mandatory.
- Warning signs shall be placed at the entrance to the test area to prevent unauthorized entry.
- **Crucial Fire Safety Note:** Prior to conducting any smoke tests, the Building Manager and/or Fire Command Centre **must be notified** to prevent accidental activation of the fire alarm system. A "smoke detector isolation" procedure may need to be followed.
- Ensure good housekeeping is maintained throughout the testing process.

# ***APPENDIX***



## ROOM RELATIVE HUMIDITY & TEMPERATURE & NOISE REPORT

|              |  |                       |  |
|--------------|--|-----------------------|--|
| PROJECT NAME |  |                       |  |
| LOCATION     |  | EQUIPMENT DESIGNATION |  |
| AREA SERVED  |  | EQUIPMENT MODEL       |  |
| SYSTEM       |  | MANUFACTURER          |  |

[illegible]

**Comments:** (If there is no additional comments enter NONE)

| Tested By   |  | Witnessed By | Verified By |
|-------------|--|--------------|-------------|
| Signature   |  |              |             |
| Name        |  |              |             |
| Designation |  |              |             |
| Company     |  |              |             |
| Date        |  |              |             |



# GRACECAL PTE LTD (GST Reg No: 201611630E)

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Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com  
Website: www.gracecal.com

## CALIBRATION CERTIFICATE

**CLIENT** : **WINNER ENGINEERING PTE LTD**  
18 Woodlands Industrial Park E1  
Singapore 757738

**CERTIFICATE NO** : GCC241442  
**ISSUE DATE** : 06-SEP-2024

**INSTRUMENT** : **MULTI FUNCTION ANEMOMETER**  
**MAKE** : TSI  
**MODEL NO.** : TA465-P-NB  
**PART NO.** : ---  
**SERIAL NO.** : TA4652029012

**WORKS ORDER NO** : GWO241601  
**DATE RECEIVED** : 02-SEP-2024

**DATE CALIBRATED** : 06-SEP-2024  
**DUE DATE** : 06-SEP-2025

**AMBIENT TEMPERATURE:**  
(23 ± 5) °C  
**RELATIVE HUMIDITY:**  
(55 ± 10) % r.h.

**RANGE** : -----  
**GRADUATION** : -----  
**(TAG NO.)** : -----  
**PAGE** : 1 of 2  
**STATUS** : AS FOUND

The described instrument has been calibrated at Laboratory under the ambient conditions stated above.

This certificate provides traceability of measurement to the International System of Units (SI) and/or to units of measurement realized at the National Metrology Centre (NMC), Singapore or other recognized national metrology institutes.

The Instrument was calibrated by comparison with laboratory calibration testing equipment.

1. Gas Deadweight Pressure Tester S/N: 558735

### Calibration Method

Calibration Method is in accordance with ISO/IEC 17025. The calibration method was carried out according to Technical Calibration Procedure AS-TM-M03 as a guide.

### Calibration Results

1. The results of calibration are given on the attached calibration data sheet(s).
2. The expanded uncertainty of measurement associated with the calibration is estimated at a confidence level of approximately 95% with a coverage factor of k=2.00.
3. The user should determine the suitability of the instrument for its intended use.

Calibrated By:

PREM

ID: 748656

Approved By:

KUMAR

ID: 748670





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## CALIBRATION CERTIFICATE

**CERTIFICATE NO** : GCC241442  
**ISSUED DATE** : 06-SEP-2024

**WORKS ORDER NO** : GWO241601  
**PAGE** : 2 of 2

| APPLIED ACTUAL VALUE (Pa)   |             | MEAN INSTRUMENT READING (Pa) |       |             |            |       |
|-----------------------------|-------------|------------------------------|-------|-------------|------------|-------|
|                             |             | Ascending                    |       | Descending  |            |       |
|                             | Before Adj. | After Adj.                   | Error | Before Adj. | After Adj. | Error |
| <b>Pressure Measurement</b> |             |                              |       |             |            |       |
| 0.00                        | 0.1         | ---                          | -0.10 | 0.0         | ---        | 0.00  |
| 50.46                       | 50.7        | ---                          | -0.24 | 50.7        | ---        | -0.24 |
| 477.53                      | 477.8       | ---                          | -0.27 | 477.8       | ---        | -0.27 |
| 916.92                      | 917.2       | ---                          | -0.28 | 917.2       | ---        | -0.28 |